

Pranav Nagarajan

Division of Physics, Math, and Astronomy
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California Institute of Technology
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Education

California Institute of Technology

Ph.D. in Astrophysics

Advisor: Kareem El-Badry

Pasadena, CA

2022-Present

GPA: 4.2 / 4.0

University of California, Berkeley

B.A. in Astrophysics, Physics, and Data Science

Advisor: Dan Weisz

Berkeley, CA

2018-2022

GPA: 4.0 / 4.0

Research Experience

Graduate Student Researcher

Advisor: Prof. Kareem El-Badry

California Institute of Technology

2022 - Present

- Discovering and Characterizing Stellar-Mass Black Holes and Neutron Stars in Galactic Binaries

Undergraduate Student Researcher

Advisor: Prof. Daniel Weisz

University of California, Berkeley

2020 - 2022

- Mapping the Local Group with RR Lyrae

BSRC Summer Internship

Mentor: Dr. Vishal Gajjar

Berkeley SETI Research Center

Summer 2021

- Breakthrough Listen Periodic Spectral Signals Search

Undergraduate Research Apprenticeship Program

Mentor: Dr. Andreas Zoglauer

Space Sciences Laboratory

2019 - 2021

- Using 3D Convolutional Neural Networks and Graph Neural Networks for Compton Track Identification in Gamma-Ray Space Telescopes

Teaching Experience

Teaching Assistant

High-Energy Astrophysics, Galaxies and Cosmology

California Institute of Technology

2024

Undergraduate Student Instructor

Principles and Techniques of Data Science

University of California, Berkeley

2020

Honors & Awards

Neugebauer Scholar	2025
NSF Graduate Research Fellowship Program: Honorable Mention	2024
Highest Distinction in General Scholarship	2022
URAP Summer Award	2020
National Merit Scholarship	2018

Awarded Telescope Time

Palomar Hale 200-inch	5 nights (PI) + 3 nights (Co-I)
Keck 10m	9 nights (Co-I)

Observing Experience

Keck II Telescope, Near-Infrared Camera 2 – 3.5 nights	2024–2025
Keck II Telescope, Near-Infrared Echelle Spectrometer – 3 nights	2024
Very Large Telescope, Focal Reducer and Low Dispersion Spectrograph 2 – 1 night	2024
Magellan I Telescope, Inamori Magellan Areal Camera and Spectrograph – 1 night	2024
Palomar Hale Telescope, Wide-Field Infrared Camera – 2 nights	2024
Keck I Telescope, Low Resolution Imaging Spectrometer – 1 night	2024
Palomar Hale Telescope, Caltech High-speed Multi-color Camera – 1 night	2023
Palomar Hale Telescope, Double Spectrograph – 3.5 nights	2023
Keck II Telescope, Echelle Spectrograph and Imager – 1 night	2022

Journal Referee

MNRAS, ApJ, AJ, OJAp, ApJL, New Astron.	6 papers total, 2023 – Present
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Talks

Talk, Binary Stars in the Space Era conference, Keele, United Kingdom	Jul. 2025
Talk, Symbiotic stars conference, Prague, Czech Republic	Jun. 2024
Tea Talk, Observatories of the Carnegie Institution for Science, Pasadena, CA	May 2024
Talk, Stellar-Mass Black Holes Group Meeting, Pasadena, CA	Jan. 2024
Talk, Zwicky Transient Facility Team Meeting, Pasadena, CA	Oct. 2023
Talk, Zwicky Transient Facility Stellar Group Meeting, Pasadena, CA	Mar. 2023

Refereed Publications

11. **Nagarajan, P.**, El-Badry, K., Chawla, C., Di Carlo, N., Breivik, K., Rodriguez, C. L., Agrawal, P., Delfavero, V., & Chatterjee, S. (2025). Realistic Predictions for Gaia Black Hole Discoveries: Comparison of Isolated Binary and Dynamical Formation Models. *Publications of the Astronomical Society of the Pacific*, 137(4), 044202.
10. Hinkle, K., **Nagarajan, P.**, Fekel, F. C., Mikołajewska, J., Straniero, O., & Muterspaugh, M. W. (2025). Binary Parameters for the Recurrent Nova T Coronae Borealis. *The Astrophysical Journal*, 983(1), 76.
9. **Nagarajan, P.** & El-Badry, K. (2025). Mixed Origins: Strong Natal Kicks for Some Black Holes and None for Others. *Publications of the Astronomical Society of the Pacific*, 137(3), 034203.
8. Blomberg, L., El-Badry K., Breivik, K., Caiazzo, I., **Nagarajan, P.**, Rodriguez, A., van Roestel, J., Vanderbosch, Z. P., & Yamaguchi, N. (2024). The Companion Mass Distribution of Post Common Envelope Hot Subdwarf Binaries: Evidence for Boosted and Disrupted Magnetic Braking? *Publications of the Astronomical Society of the Pacific*, 136(12), 124201.
7. **Nagarajan, P.** & El-Badry, K. (2024). Validation of Gaia DR3 orbital and acceleration solutions with hierarchical triples. *Publications of the Astronomical Society of the Pacific*, 136(9), 094203.
6. **Nagarajan, P.**, El-Badry, K., Lam, Y. C., & Reggiani, H. (2024). The Symbiotic X-ray Binary IGR J16194-2810: A Window on the Future Evolution of Wide Neutron Star Binaries From Gaia. *Publications of the Astronomical Society of the Pacific*, 136(7), 074202.
5. **Nagarajan, P.**, El-Badry, K., Triaud, A. H., Baycroft, T. A., Latham, D., Bieryla, A., Buchhave, L. A., Rix, H.-W., Quataert, E., Howard, A., Isaacson, H., & Hobson, M. J. (2024). ESPRESSO Observations of Gaia BH1: High-precision Orbital Constraints and no Evidence for an Inner Binary. *Publications of the Astronomical Society of the Pacific*, 136(1), 014202.
4. **Nagarajan, P.**, El-Badry, K., Rodriguez, A. C., van Roestel, J., & Roulston, B. (2023). Spectroscopic follow-up of black hole and neutron star candidates in ellipsoidal variables from *Gaia* DR3. *Monthly Notices of the Royal Astronomical Society*, 524(3), 4367–4383.
3. El-Badry, K., Shen, K. J., Chandra, V., Bauer, E. B., Fuller, J., Strader, J., Chomiuk, L., Naidu, R. P., Caiazzo, I., Rodriguez, A. C., **Nagarajan, P.**, Yamaguchi, N., Vanderbosch, Z. P., Roulston, B. R., Gänsicke, B., Han, J. J., Burdge, K. B., Filippenko, A. V., Brink, T. G., & Zheng, W. (2023). The fastest stars in the galaxy. *The Open Journal of Astrophysics*, 6.
2. Suresh, A., Gajjar, V., **Nagarajan, P.**, Sheikh, S. Z., Siemion, A. P., Lebofsky, M., MacMahon, D. H., Price, D. C., & Croft, S. (2023). A 4–8 GHz Galactic Center Search for Periodic Technosignatures. *The Astronomical Journal*, 165(6), 255.
1. **Nagarajan, P.**, Weisz, D., & El-Badry, K. (2022). RR Lyrae-based distances for 39 nearby dwarf galaxies calibrated to *Gaia* EDR3. *The Astrophysical Journal*, 932(1), 19.